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Submitted by email to [REDACTED]

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Re: Request for Information on the Development of an AI Action Plan

Mr. D'Souza:

Thank you for the opportunity to submit a comment regarding the United States' upcoming AI Action Plan. I have been following developments in the AI field closely for more than a decade now, and while I am not a US resident, the Action Plan will clearly have an impact on me, and the world outside of the US.

Regards,

Myron Hedderson

Introduction

This submission outlines strategic recommendations for the AI Action Plan to ensure Western leadership in artificial intelligence while addressing critical security challenges. As a cybersecurity specialist with over 15 years of experience, including work protecting critical infrastructure at NB Power (a Canadian electric utility) from potential nation-state threats since 2021, I can provide a unique perspective on the challenges of securing AI systems from determined adversaries. I have been following developments in AI closely since 2014, giving me insight into both the cybersecurity and AI domains.

In this submission, I propose a framework that supports robust innovation and technological primacy while implementing targeted safeguards against the most severe risks. My recommendations are designed to enhance the United States' strategic advantage and allow the US and the Western world to take full advantage of potentially world-changing AI capabilities, while preventing potential catastrophic vulnerabilities that could undermine national security and democratic values.

The pace of AI development is far faster than government processes are designed to respond to. Over the past 2 years, AI systems have moved from being a surprisingly good chatbot, to outperforming doctors on diagnosis, generating tens of thousands of candidate bioweapons in a day, and in general approaching human expert level in many domains. Within the current Presidential term, frontier AI companies project that their models will move to and likely past human capabilities in all domains. This compressed timeline means decisions made now will determine whether the US maintains its leadership position or cedes ground to competitors. The window for establishing effective governance while maintaining our technological edge is rapidly closing.

Advanced AI systems represent both the greatest technological opportunity of our time and, if developed incorrectly, a potential strategic vulnerability. With a Chinese Communist Party (CCP) determined to overtake Western democracies by 2030, the Trump Administration's AI Action Plan can ensure that democratic values remain embedded in AI development. The recommendations in this submission will help ensure that AI systems remain aligned with democratic principles and interests, preventing scenarios where these powerful technologies could act against our shared objectives.

A Perspective from Cybersecurity: The Challenge of Protecting AI Assets

The protection of critical AI assets – particularly the model weights of frontier AI systems – presents unprecedented security challenges:

1. **Scale and Complexity:** Advanced AI models represent some of the most complex software systems ever created, with billions or trillions of parameters. This complexity creates an expansive attack surface that is difficult to fully secure.
2. **Attractive Target:** Model weights represent billions of dollars in R&D investment and computational resources. This makes them extremely high-value targets for theft by sophisticated adversaries like the CCP.
3. **Transferable Value:** Unlike many cybersecurity targets where the value diminishes once stolen, model weights retain their full value and capabilities even after being compromised.
4. **Nation-State Resources:** Nation-state adversaries can maintain persistent access to even the most secured systems, as demonstrated by the case study in recommendation 3. This level of capability makes protecting model weights exceptionally challenging.

Despite, but informed by, my expertise and long experience in Cybersecurity, if asked to protect a critical asset from a determined nation-state attacker, my response would be “we can make their task harder, but I can’t give you a guaranteed way of keeping adversaries out - the attempt to gain access will likely succeed in time”. I believe we must adopt a security posture that assumes breaches will occur and focuses on limiting the damage when they do. This requires defense-in-depth approaches, robust monitoring, and the development of technical measures that can render stolen models less valuable to adversaries. It is also noteworthy that we do not currently have technical measures that can reliably prevent someone who has access to the weights of a model from accessing all of its capabilities. Even without access to model weights, new versions of frontier models are typically “jailbroken” within days of release. Given the pace of recent advances, developing measures to reliably assess and control what AI models can and cannot do should be a key research priority. Without solving that problem, there will soon come a point where the choice is between deploying an advanced model that has a significant risk of causing catastrophic harm, or stopping deployment of more advanced models, and taking fairly extreme measures in the name of national security to enforce that moratorium.

Recommendations

1. Safeguard Against Catastrophic Vulnerabilities

Recommendation: Establish a dedicated program within the Department of Defense or intelligence community focused on identifying and mitigating potential catastrophic security vulnerabilities that could be created by advanced AI systems.

Rationale: Advanced AI could potentially be used to identify novel cybersecurity vulnerabilities, develop biological threats, or create other security risks that could threaten national security. Additionally, agent-based or autonomous AI systems could take actions that pose national security threats. Preventing such scenarios is essential to national defense in the AI age and ensuring that adversaries, such as the CCP, cannot use AI to gain asymmetric advantages. Although the threats that AI may pose are widely recognized, appropriate countermeasures or control strategies are not yet available.

If advanced AI systems were to fall into adversarial hands or develop dangerous capabilities, the national security implications could be severe. This program would develop both preventative measures and response capabilities for such scenarios.

Implementation:

- 1.1 Create specialized red teams specifically tasked with identifying potential ways advanced AI could threaten critical infrastructure or national security
- 1.2 Develop contingency plans and technical countermeasures for identified risks
- 1.3 Establish protocols for emergency interventions if critical capability thresholds are crossed without appropriate control measures.
- 1.4 Implement continuous monitoring systems for detecting potential misuse of advanced AI capabilities.
- 1.5 Develop technical countermeasures that could rapidly neutralize AI systems that demonstrate dangerous capabilities
- 1.6 Create secure communications channels for reporting critical AI vulnerabilities without public disclosure.
- 1.7 Conduct regular tabletop exercises simulating AI security crises to test response capabilities. These exercises should in at least some cases be whole-of-government and involve the leading AI companies, rather than being limited to the defense or intelligence communities.
- 1.8 Track CCP-developed AI systems for potential dual-use capabilities that could be weaponized against democratic nations
- 1.9 Provide classified threat briefings to AI developers to help them defend against sophisticated nation-state threats
- 1.10 Develop collaborative approaches with trusted allies like Canada, who share intelligence through frameworks like Five Eyes

- 1.11 Require binding security commitments from major AI developers, with regular independent audits to verify compliance

2. Increase Federal Funding and Support for AI Alignment and Control Efforts

Recommendation: Significantly increase federal funding for research into technical AI safety measures that ensure advanced AI systems remain firmly under human control and aligned with democratic values. This could be implemented through the newly established US AI Safety Institute, working in coordination with DARPA, the National Science Foundation, and relevant intelligence community stakeholders.

Rationale: Technological advantage must include not just raw AI capabilities but also the ability to ensure these systems remain under control - the freedom to build must be matched with the freedom to direct these systems as intended. A system that cannot be controlled is no advantage at all. The Grok incident in early 2025, where the system developed by Elon Musk's X recommended death for American leaders including President Trump, demonstrates the importance of ensuring AI systems' outputs align with core democratic values, and the fact that currently understood technical measures are not sufficient for this purpose.

Implementation:

- 2.1 Establish a dedicated research program modeled after DARPA focusing specifically on AI alignment and control technologies
- 2.2 Prioritize funding for research into methods that prevent AI systems from developing capabilities that could threaten strategic interests, or neutralizing specific capabilities within a broadly-trained model. These should include:
 - techniques for interpretability of model decision-making
 - reliable containment mechanisms for testing potentially dangerous capabilities
 - technical measures to prevent model theft or unauthorized use
 - robust methods for detecting and preventing unauthorized self-improvement.
- 2.3 Create economic incentives for private sector companies to develop and implement robust safety measures. This could include R&D tax credits specifically for AI safety research, regulatory safe harbors for companies implementing approved safety measures, and prioritized access to government contracts for those demonstrating best-in-class safety protocols.
- 2.4 Fund a "NASA-like" initiative for AI safety that unites the best minds to solve the most significant challenges in maintaining control of increasingly powerful AI systems
- 2.5 Position Western democracies as world leaders in not just AI capabilities but in AI control mechanisms, creating potential for export of safety technologies to allied nations

3. Leverage Compute Advantage While Addressing Critical Security Vulnerabilities

Recommendation: Develop a strategic approach to compute resources that maintains the United States' hardware advantage while implementing robust security measures to protect AI infrastructure and model weights from foreign compromise.

Rationale: Computing infrastructure represents a clear advantage in the global AI race. The hardware required for training the most advanced AI systems is predominantly designed and controlled by Western companies. However, running an AI model takes much less compute than training one, so control of critical chip supply chains is most valuable when AI model weights are also well protected. At the moment, the most advanced AI models are in the hands of private companies with commercial incentives which may lead to some shortcuts on security. However, the US also leads in cybersecurity, with many of the foremost experts in this area residing within the NSA or National Defense establishment. The skills and capabilities of these experts should be made available to leading AI companies, to strengthen their position relative to foreign rivals.

A Case Study: Salt Typhoon

The recent Salt Typhoon cyber operation against multiple U.S. telecommunications providers offers important lessons for AI security.

As reported by various sources¹, Chinese hackers maintained access to the networks of at least 9 major US telecommunications providers for over a year, compromising call records of over a million users (including government officials, presidential campaign staff, and the President and Vice President), and even gaining visibility into lawful wiretapping systems. This incident demonstrates several critical points relevant to AI security:

1. **Even the most protected systems remain vulnerable** - Despite substantial cybersecurity investments, telecommunications networks central to national security were compromised for an extended period. Admittedly, telecommunications networks present a very large attack surface and are difficult to secure, but AI models that are widely deployed would have a similarly large attack surface, and even airgapped networks are vulnerable to attack from a determined adversary, as demonstrated by the NSA's hack of Iranian centrifuges with the Stuxnet worm.
2. **Nation-state adversaries possess advanced capabilities** - The sophistication and persistence of this attack demonstrates the resources and determination that nation-state actors like the CCP bring to targeting critical Western technology.

¹ https://en.wikipedia.org/wiki/2024_United_States_telecommunications_hack

3. **Strategic intelligence is compromised** - The breach of lawful interception systems demonstrates how cybersecurity failures can directly impact intelligence capabilities.
4. **Recovery is difficult and time-consuming** - Despite months of remediation efforts, complete eviction of the attackers has proven challenging.

The implications for AI security are profound. If we cannot fully secure telecommunications infrastructure against determined nation-state adversaries, we must approach AI security with even greater vigilance. Advanced AI systems represent potentially more attractive targets that could provide adversaries with even greater strategic advantages if compromised.

Implementation:

- 3.1 Continue and enhance existing export controls on advanced AI chips to countries of concern, particularly the CCP.
- 3.2 Develop contingency measures to prevent the use of computational resources for foreign AI development that could threaten national security. This likely includes tracking chip supplies, and being able to disable chips if needed.
- 3.3 Create incentives for construction of additional AI compute infrastructure within democratic nations, including through permitting reform and energy infrastructure investment.
- 3.4 Collaborate with leading AI companies to establish a specialized cybersecurity framework specifically designed for AI infrastructure protection.
- 3.5 Develop secure enclaves for the most sensitive AI training operations with air-gapped networks and enhanced physical security.
- 3.6 Implement advanced threat detection systems specifically designed to identify attempts to exfiltrate model weights or training data.
- 3.7 Create redundant security architectures assuming that perimeter defenses will be breached (defense-in-depth approach).
- 3.8 Consider creating specialized AI Economic Zones with fast-track permitting for AI infrastructure and enhanced security measures
- 3.9 Build cooperation between major AI centers in North America, leveraging both American industrial scale and Cybersecurity expertise

4. Create an Advanced AI Testing & Evaluation Center

Recommendation: In collaboration with leading AI companies and existing testing organizations such as [METR](#), establish a national center for testing and evaluating advanced AI systems, both during training and post-deployment, to ensure they perform as intended and do not contain hidden vulnerabilities or capabilities that could be exploited by adversaries or pose a national security risk.

Rationale: Strategic advantage requires that AI systems work as designed and do not contain unexpected behaviors when deployed. A robust testing infrastructure will ensure that AI systems maintain this advantage and do not create unexpected security vulnerabilities. At

present, the leading experts in AI testing reside within the leading AI companies and a few non-government testing organizations. While there should be standards for appropriate testing, my experience with government-based standards bodies such as NIST (who creates cybersecurity standards, among many other things) is that their standards take several years to develop, and are often sufficiently vague as to allow a wide variety of interpretations. As the CCP attempts to catch up to Western AI capabilities, testing and evaluation standards can become another area where democracy-aligned values set the pace for secure, reliable AI – but they have to be specific enough to be useful, and developed and updated on a timeline that recognizes how quickly AI is advancing.

Implementation:

- 4.1 Create a dedicated facility with appropriate security measures and computational resources for testing advanced systems, and testing the most sensitive capabilities without risking exposure of test methodologies to potential adversaries
- 4.2 Develop standardized testing protocols that can identify potential security and control issues
- 4.3 Offer voluntary pre-deployment evaluations to US AI developers in a manner that protects intellectual property while identifying critical safety issues. Aim to have these tests in place quickly, so that the testing process will have matured somewhat by the time models that introduce significant national security problems are being tested.
- 4.4 For models where there are reasonable grounds to believe they may pose wide-scale societal harms, or pose a national security threat if model weights are in the hands of an adversary, require evaluation for critical safety issues.
- 4.5 Implement adversarial testing methodologies specifically designed to identify potential security vulnerabilities, harmful outputs, or control failure.
- 4.6 Develop specialized instrumentation for monitoring AI system behavior during testing that can identify subtle signs of emerging problematic capabilities
- 4.7 Establish protocols for responsible disclosure of identified vulnerabilities to developers without creating public security risks
- 4.8 Coordinate with allied nations to develop shared testing standards that could become de facto global norms

5. Develop an AI Safety/Security Workforce

Recommendation: Establish a comprehensive national initiative to develop the specialized workforce needed to secure AI systems against both technical failures and adversarial threats.

Rationale: The security of AI systems will ultimately depend on having sufficient human expertise to identify, prevent, and respond to potential security threats. Current cybersecurity workforce shortages will be exacerbated as AI systems grow more complex and require specialized security knowledge. The United States cannot maintain its AI leadership without a world-class workforce that can secure these systems against sophisticated adversaries like the CCP.

Implementation:

- 5.1 Create specialized training programs at federal institutions (including military academies and service academies) focused on AI security
- 5.2 Establish AI security fellowships to attract top talent to government service.
- 5.3 Fund academic research centers focused specifically on AI security challenges
- 5.4 Develop specialized career tracks within key agencies (DOD, intelligence community, DHS) for AI security professionals
- 5.5 Create executive education programs to ensure leadership across government and industry understand AI security fundamentals
- 5.6 Establish rapid response teams with specialized AI security expertise that can be deployed in crisis situations
- 5.7 Create streamlined immigration pathways for AI security experts from close allies like Canada to bring their talents to the United States.
- 5.8 Explore immigration pathways for experts from non-allied nations. Where appropriate security measures are taken to ensure loyalty to democratic values, immigration pathways for experts who would like to leave a repressive regime would be a win for the US and a loss for our adversaries
- 5.9 Create tax incentives for businesses investing in AI security workforce development

Supporting Innovation While Mitigating Strategic Risks

These recommendations are designed to support innovation and technological leadership while preventing scenarios that could undermine national security. Rather than imposing blanket regulations that would hinder America's competitive edge, this approach focuses on targeted interventions that address the most severe potential risks, with binding requirements in areas where the risks are greatest.

The proposed measures would:

- 1 **Enhance technological leadership** by ensuring the West maintains its edge over competitors.
- 2 **Protect national security** by preventing catastrophic vulnerabilities and threats.
- 3 **Maintain democratic values** by ensuring AI systems act in accordance with shared interests and democratic principles.
- 4 **Support private sector innovation** by focusing on targeted requirements rather than overly restrictive regulation.
- 5 **Build critical security infrastructure** by developing both the technical systems and human expertise needed to secure advanced AI.
- 6 **Create economic opportunities** by positioning democracies as world leaders in both AI capabilities and AI security technologies.

Conclusion

The United States and its democratic allies stand at the forefront of the global AI revolution, with a Chinese Communist Party determined to overtake the West by 2030. The recommendations outlined in this submission will help ensure that democratic values prevail in AI development while preventing potential catastrophic outcomes that could undermine our technological advantage. By implementing targeted measures to address the most severe risks, Western democracies can continue to lead in AI innovation while ensuring these technologies remain firmly aligned with democratic interests and values.

I appreciate the opportunity to submit these recommendations and would be happy to provide additional information or clarification as needed.